

Sourasish Ghosh

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PROFILE

BS in Data Science and M.Sc. in Atmospheric Science with a strong foundation in Data Science and Physics. Skilled in **ML/DL, statistical modeling, and data engineering**, with hands-on experience building and deploying **end-to-end ML systems** (DNNs, Random Forests, Fuzzy models).

WORK EXPERIENCE

ML and System Engineer

Feb 2026 - Current

Spaceborn

- Architected a real-time ROS 2 obstacle avoidance stack, implementing an Artificial Potential Field (**APF**) local planner, Unscented Kalman Filter (**UKF**) object tracking, and **YOLO**-based perception to ensure safe, 50Hz autonomous navigation.
- Engineered an anomaly detection engine for high-frequency flight telemetry utilizing **Isolation Forest**, enabling real-time identification of sensor degradation and erratic UAV behavior.
- Designed a bi-directional cloud-to-edge telemetry bridge using **Apache Kafka** and **Redis**, implementing Avro binary serialization to minimize bandwidth latency between the drone and a **FastAPI backend**.
- Fused multi-modal sensor data (TF-Luna LIDAR, HC-SR04 sonar) through a dedicated ESP32 co-processor.
- Integrated custom ROS 2 AI nodes with standard Pixhawk/MAVLink flight controllers via MAVROS.

Data Analyst Intern

Dec 2025 - Jan 2025

Excelerate

- Conducted analysis on Excelerate's datasets for their candidate hirings as a **Project Lead** of our team.
- Used **BI**-tools to visualize and for analysis used **Python, Excel**.
- Caught key-findings in **Yield management, Geograohic dominance** of canidate approach of nearly 200 Countries, **Efficiency** problems etc.
- Helped building **KPIs** of the business with future recommendation.

Workshop: Introduction to Cloud using GCP

May 2025

Indian Institute of Technology, Madras (Online)

[Link](#)

- Worked with **Compute Engine, BigQuery, Cloud Storage, Cloud Functions, Pub/Sub**.
- Built a **dashboard system** for real-time cloud-based data streams using **Pub/Sub** and **Cloud Functions**.

Microsoft Azure Certifications

[Link](#)

- Introduction to Cloud Infrastructure (AZ-900 series)

PROJECTS

Python-based Finance Backend System

(2026)

- Engineered a high-performance financial backend using **FastAPI** and **SQLAlchemy**, implementing a modular architecture with a strict separation of concerns (**Routers, CRUD, Models, Schemas**).
- Designed and managed a relational schema in **PostgreSQL**, utilizing **Pydantic** for data validation and type safety across the application lifecycle.
- Role-Based Access Control (**RBAC**): Implemented a multi-tier authorization system (Admin, Analyst, Viewer) to secure sensitive financial data and restrict endpoint access based on user privileges.

Coughometer – DNN-based Cough Classifier

(2025)

- Built an end-to-end app with **DNN**, FastAPI backend, HTML/JS frontend.
- Packaged with Docker and deployed on **Render**.
- Trained using public datasets (**Coswara, COUGHVID**).

Thesis: Hybrid Fuzzy–Random Forest Model for Severe Weather Prediction

(2025)

- Developed a hybrid model integrating **Fuzzy Logic + Random Forest**.
- Achieved **88.6% accuracy** on 24 years Kolkata weather data.

BDM: Inventory Optimization for a local Café Business

(2025)

- A Capstone-Project under IITM about optimizing inventory and maximizing profit of a local Cafe.
- Performed SKU-level analysis, regression ($R^2 \approx 0.96$), and forecasting.

TECHNICAL SKILLS

Languages: **Python, Fortran, SQL, $\text{\LaTeX}$** ; *Cloud / DB:* **GCP, PostgreSQL, Render**

Libraries: **NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, Tensorflow, PyTorch, Librosa, TabNet, PySpark**

Tools: **FastAPI, Docker, Flask, NetCDF, QGIS, Apache Kafka, Redis** ; *Environments:* **Linux/Unix, VS Code, Git, Github**

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management, Feature Engineering, Meteorological Data Interpretation, Business Analytics, Statistic, NLP, Machine Learning, A/B testing, Statistical Inference.

EDUCATION

Master of Science in Atmospheric Sciences

University of Calcutta, Kolkata

(2023 — 2025)

CGPA: 8.155

Bachelor of Science in Data Science and Applications (*Distance*)

Indian Institute of Technology, Madras

(Concurrent with Master of Science in Atmospheric Science)

(2023 – 2027)

CGPA: 7.67

Bachelor of Science in Physics (Honours)

Chandernagore College, University of Burdwan

(2020 — 2023)

CGPA: 7.72